UNIVERSITY OF WAH(UOW)

DEPARTMENT OF COMPUTER SCIENCE

Competence through Excellence
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UNIVERSITY OF WAH

Lab Task # 05

COURSE: Data Structure & Algorithm Lab_201L

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Task # 01

```
int arr[] = {2,4,6,8,1,9,14};
int n = 7;
int key, low = 0, high = n-1, mid;
cout<<"Enter Number to Search: ";
cin>>key;
while (low <= high)
{
    mid = (low + high)/2;
    if (arr[mid] == key)
    {
        cout<<"\nFound at index "<<mid<<endl;
        return 0;
    }
    else if (key < arr[mid])
    {
        high = mid - 1;
    }
    else{
        low = mid + 1;
    }
}
cout<<"Number not found."<<endl;
return 0;
```

Output

```
Enter Number to Search: 14

Found at index 6
```

Task # 02

```
int arr[5] = {3, 1, 5, 2, 4};
int n = 5;
for (int i = 0; i < n - 1; i++)
{
    for (int j = 0; j < n - i - 1; j++)
    {
        if (arr[j] > arr[j + 1])
        {
            swap(arr[j],arr[j+1]);
        }
    }
}
for (int i = 0; i < n; i++)
{
    cout << arr[i] << " ";
}
```

Output

```
PS D:\University Data\3rd
1 2 3 4 5
```

Task # 03

```
void swapNumbers(int* ptr1, int* ptr2) {

    int temp = *ptr1;
    *ptr1 = *ptr2;
    *ptr2 = temp;
}

int main() {
    int num1 = 10;
    int num2 = 20;

    cout << "Before swapping: num1 = " << num1 << ", num2 = " << num2 << endl;
    swapNumbers(&num1, &num2);
    cout << "After swapping: num1 = " << num1 << ", num2 = " << num2 << endl;

    return 0;
```

Output

```
Before swapping: num1 = 10, num2 = 20
After swapping: num1 = 20, num2 = 10
```

Task # 04

```cpp
struct Employee {
    string name;
    int emp_id;
    float salary;
};

int main() {
    Employee emp[3], temp;
    string searchName;
    bool found = false;
    cout << "Enter data for 3 employees:\n";
    for (int i = 0; i < 3; i++) {
        cout << "\nEmployee " << i + 1 << ":\n";
        cout << "Name: ";
        cin >> emp[i].name;
        cout << "Emp ID: ";
        cin >> emp[i].emp_id;
        cout << "Salary: ";
        cin >> emp[i].salary;
    }
    for (int i = 0; i < 3 - 1; i++) {
        for (int j = 0; j < 3 - i - 1; j++) {
            if (emp[j].salary < emp[j + 1].salary) {
                temp = emp[j];
                emp[j] = emp[j + 1];
                emp[j + 1] = temp;
            }
        }
    }
```

```cpp
    cout << "\nEmployees sorted by highest salary:\n";
    for (int i = 0; i < 3; i++) {
        cout << "\nName: " << emp[i].name
             << "\nEmp ID: " << emp[i].emp_id
             << "\nSalary: " << emp[i].salary << endl;
    }
    cout << "\nEnter employee name to search: ";
    cin >> searchName;

    for (int i = 0; i < 3; i++) {
        if (emp[i].name == searchName) {
            found = true;
            cout << "\nEmployee Found!\n";
            cout << "Name: " << emp[i].name
                 << "\nEmp ID: " << emp[i].emp_id
                 << "\nSalary: " << emp[i].salary << endl;
            break;
        }
    }
```

```cpp
    }
    if (!found) {
        cout << "\nEmployee with name '" << searchName << "' not found.\n";
    }
```

Output

```
Employees sorted by highest salary:

Name: Ehmad
Emp ID: 333
Salary: 500

Name: Talha
Emp ID: 222
Salary: 200

Name: Danish
Emp ID: 123
Salary: 100

Enter employee name to search: Talha

Employee Found!
Name: Talha
Emp ID: 222
Salary: 200
```

Task

```cpp
#include <iostream>
#include <cstring>
using namespace std;

struct Roll {
    char department[20];
    char session[20];
    int registration;
    char degree[20];
};
struct Student {
    char name[50];
    int age;
    Roll roll;
    float marks[3];
    float gpa;
};
```

```cpp
void displayStudent(Student s) {
    cout << "------------------------------------\n";
    cout << "Name: " << s.name << endl;
    cout << "Age: " << s.age << endl;

    cout << "Department: " << s.roll.department << endl;
    cout << "Session: " << s.roll.session << endl;
    cout << "Registration No: " << s.roll.registration << endl;
    cout << "Degree: " << s.roll.degree << endl;

    cout << "Marks: " << s.marks[0] << ", "
         << s.marks[1] << ", "
         << s.marks[2] << endl;

    cout << "GPA: " << s.gpa << endl;
    cout << "------------------------------------\n";
}
```

```cpp
int binarySearch(Student s[], int size, int key) {
    int left = 0;
    int right = size - 1;

    while (left <= right) {
        int mid = (left + right) / 2;

        if (s[mid].roll.registration == key)
            return mid;
        else if (key < s[mid].roll.registration)
            right = mid - 1;
        else
            left = mid + 1;
    }
    return -1;
}
```

```cpp
int main() {
    Student st[5];
    cout << "Enter data of 5 students:\n";
    for (int i = 0; i < 5; i++) {
        cin.ignore();

        cout << "\nEnter name: ";
        cin.getline(st[i].name, 50);

        cout << "Enter age: ";
        cin >> st[i].age;
```

```cpp
    cin.ignore();
    cout << "Enter department: ";
    cin.getline(st[i].roll.department, 20);

    cout << "Enter session: ";
    cin.getline(st[i].roll.session, 20);

    cout << "Enter registration number: ";
    cin >> st[i].roll.registration;

    cin.ignore();
    cout << "Enter degree: ";
    cin.getline(st[i].roll.degree, 20);

    cout << "Enter 3 subject marks: ";
    cin >> st[i].marks[0] >> st[i].marks[1] >> st[i].marks[2];

    st[i].gpa = (st[i].marks[0] + st[i].marks[1] + st[i].marks[2]) / 3.0;
}
```

```cpp
    for (int i = 0; i < 5 - 1; i++) {
        for (int j = 0; j < 5 - i - 1; j++) {
            if (st[j].gpa < st[j + 1].gpa) {
                Student temp = st[j];
                st[j] = st[j + 1];
                st[j + 1] = temp;
            }
        }
    }

    cout << "\n\n===== STUDENTS SORTED BY GPA (Highest First) =====\n";
    for (int i = 0; i < 5; i++) {
        displayStudent(st[i]);
    }

    int key;
    cout << "\nEnter registration number to search: ";
    cin >> key;

    int index = binarySearch(st, 5, key);

    if (index != -1) {
        cout << "\nStudent Found!\n";
        displayStudent(st[index]);
    } else {
        cout << "\nStudent NOT found!\n";
    }
    return 0;
```

Output

```
===== STUDENTS SORTED BY GPA (Highest First) =====
------------------------------------
Name: Fahad
Age: 90
Department: CS
Session: 2021
Registration No: 12
Degree: AI
Marks: 90, 89, 76
GPA: 85
------------------------------------
------------------------------------
Name: Ehmad
Age: 67
Department: CS
Session: 2012
Registration No: 898
Degree: DS
Marks: 89, 68, 54
GPA: 70.3333
------------------------------------
------------------------------------
Name: Osama
Age: 65
Department: Raw
Session: 2021
Registration No: 902
Degree: CS
Marks: 23, 45, 65
GPA: 44.3333
------------------------------------
```

```
------------------------------------
Name: Rafay
Age: 90
Department: CS
Session: 2021
Registration No: 202
Degree: DS
Marks: 34, 54, 34
GPA: 40.6667
------------------------------------
------------------------------------
Name: ALI
Age: 89
Department: DS
Session: 2012
Registration No: 1111
Degree: CA
Marks: 23, 43, 54
GPA: 40
------------------------------------

Enter registration number to search: 45

Student NOT found!
```